

[← Go Back](#)

Hardware

Nano 33 BLE Sense Rev2 ▾

Tutorials

Access Barometric
Pressure Sensor Data on
Nano 33 BLE Sense

Connecting Nano 33 BLE
Devices over Bluetooth®

Controlling RGB LED
Through Bluetooth®

Nano 33 BLE Sense Rev2
User Manual

Nano 33 BLE Sense
Community Projects

Edge Impulse with the
Nano 33 BLE Sense

Gesture Recognition with
the Nano 33 BLE Sense

Get Started With Machine
Learning on Arduino

Getting Started with
OpenMV

Reading Temperature &
Humidity on Nano 33 BLE
Sense Rev2

Connecting Two Nano 33
BLE Sense Boards Through
I2C

Accessing Accelerometer
Data on Nano 33 BLE Sense
Rev2

Accessing Gyroscope Data
on Nano 33 BLE Sense Rev2

Accessing Magnetometer
Data on Nano 33 BLE Sense
Rev2

Home / Hardware / Nano 33 BLE
Sense Rev2 / **Access Barometric
Pressure Sensor Data on Nano 33
BLE Sense**

Access Barometric Pressure Sensor Data on Nano 33 BLE Sense

Learn how to read data from the
LPS22HB barometric pressure sensor
on the Nano 33 BLE Sense board.

Author · Nefeli Last
Alushi revision · 10/31/2025

ON THIS PAGE

Goals

Hardware & Software Needed

LPS22HB Sensor +

Atmospheric Pressure and
Altitude

Creating the Program

Testing It Out +

Conclusion

This tutorial will use an Arduino NANO 33 BLE Sense, to calculate the approximate altitude above sea level through the measurement of the atmospheric pressure, made possible by the embedded **LPS22HB** sensor.

A popular application of a barometric sensor, apart from GPS and forecasting short term changes in the weather, is the altitude detection according to the atmospheric pressure. This tutorial will focus on the conversion of the atmospheric pressure data, in order to approximately identify the altitude values of the user's surrounding environment.

Goals

The goals of this project are:

- Learn what a LPS22HB sensor is.

- Use the [LPS22HB library](#).

- Learn how to output raw sensor data from the Arduino NANO 33 BLE Sense.

- Learn how to convert the kPa unit to altitude values (meters).

- Print the data using Serial Monitor.

- Create your own barometer that reveals your location's altitude.

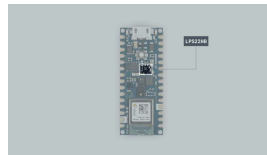
Hardware &

This project uses no external sensors or components.

In this tutorial we will use the [Arduino Create Cloud Editor](#) to program the board.

LPS22HB Sensor

The LPS22HB is an ultra-compact piezoresistive absolute pressure sensor which functions as a digital output barometer. The device comprises a sensing element and an IC interface which communicates through I2C or SPI, from the sensing element to the application.



The LPS22HB sensor.

The sensing element, which detects absolute pressure, consists of a suspended silicon membrane and it operates over a temperature range extending from -40 °C to +85 °C. The functionality of the sensor will be explained further on later on the tutorial.

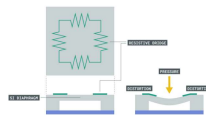
If you want to read more about the LPS22HB sensor module see [here](#).

The Library

sensor of the Nano 33 BLE Sense, without having to go into complicated programming. The library also takes care of the sensor initialization with the function `BARO.begin()` as well as de-initializes the sensor with the function `BARO.end()`.

Atmospheric Pressure and Altitude

Modern day barometers, known as barometric pressure sensors contain a diaphragm that's formed through a resistive plate that's in contact with the atmosphere as seen in the image below.



How atmospheric pressure and altitude works.

The atmospheric pressure is detected according to how much the diaphragm has deformed, due to resulting pressure. The higher the pressure is, the more the diaphragm moves, which result in a higher barometer reading.

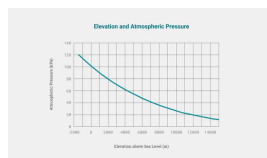
Once we retrieve the sensor values of the atmospheric pressure in kPa (unit of

to calculate the environment's approximate altitude in meters:

$$H = 44330 * [1 - (P/p_0)^{(1/5.255)}]$$

Where, "**H**" stands for altitude, "**P**" the measured pressure (kPa) from the sensor and "**p₀**" is the reference pressure at sea level (101.325 kPa).

This graph shows the mathematical relationship between atmospheric pressure and elevation above sea level.

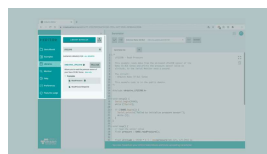


Elevation and atmospheric pressure graph.

Creating the Program

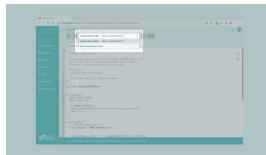
1. Setting up

Let's start by opening the [Arduino Cloud Editor](#), click on the **Libraries** tab and search for the **LPS22HB** library. Then in > **Examples**, open the **ReadPressure** sketch and once it opens, rename it as **Barometer**.



2. Connecting the board

Now, connect the Arduino Nano 33 BLE Sense to the computer and make sure that the Cloud Editor recognizes it, if so, the board and port should appear as shown in the image below. If they don't appear, follow the [instructions](#) to install the plugin that will allow the Editor to recognize your board.



Selecting the board and port.

3. Calculating the approximate altitude

Now we will need to modify the code on the example, in order to print the board's approximate altitude in meters, through the measurement of the barometric sensor.

After including the

`Arduino_LPS22HB.h` library the

`setup()` section will remain

the same. Here, the function initializes the Serial

communication at a standard rate of 9600 bauds, as well as the sensor. In case the sensor has failed to initiate, an error message will be printed.

In the `loop()` the function

`BARO.readPressure()` retrieves

the live data from the sensor

need to tweak and add the aforementioned mathematical formula to convert the pressure sensor data into altitude:

```
float altitude = 44330 * (
  1 - pow(pressure/101.325,
    1/5.255) );
```

In this formula we simply change the formatting and converted the equations into code.

Note: The pow() function computes a base number raised to the power of exponent number.

Lastly, we adapt the `serial.println()` functions to the data we want to print, which is altitude in meters:

```
1 Serial.print("Altitude")
2 Serial.print(altitude)
3 Serial.println(" m");
```

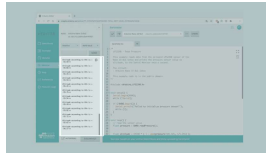
4. Complete code

If you choose to skip the code building section, the complete code can be found below:

```
1  /*
2   LPS22HB - Read Pressure and Temperature
3
4   This example reads the LPS22HB sensor,
5   converts the atmospheric pressure to altitude,
6   and prints them to the Serial Monitor.
7
8   The circuit:
9   - Arduino Nano 33 BLE
10
11   This example code is in the public domain.
12  */
13
14  #include <Arduino_LPS22HB.h>
15
16
17  void setup() {
18    Serial.begin(9600);
19    while (!Serial);
20
21    if (!BARO.begin()) {
22      Serial.println("Error: BARO not found");
23      while (1);
24    }
25  }
26
27  void loop() {
28    // read the sensor
```

Testing It Out

After you have successfully verified and uploaded the sketch to the board, open the Serial Monitor from the menu on the left. In order to test out the code, you could begin by stabilizing your board on a fixed position and observe the values returned through the Serial Monitor. If you are living in an apartment block, you could experiment further and move to different floors and notice the changes in altitude values.



Pressure data printed
in the Serial Monitor.

Troubleshoot

Sometimes errors occur, if the code is not working there are some common issues we can troubleshoot:

- Missing a bracket or a semicolon.

- Arduino board connected to the wrong port.

- Accidental interruption of cable connection.

Conclusion

In this tutorial we learned what a **LPS22HB** sensor is, how to use the one embedded in the Arduino NANO 33 BLE Sense board and the LPS22HB library, in order to create our own barometer and measure the altitude in meters by retrieving and calculating the atmospheric pressure.

Suggest changes Need support?

The content on docs.arduino.cc is

[Help Center](#)
[Ask the Arduino Forum](#)

License

The Arduino documentation is licensed under the [Creative Commons](#)

GitHub
repository.
If you see
anything
wrong, you
can edit
this page
[here](#).

[Share Alike](#)
[4.0](#) license.

Was this article helpful?

